



Department of
Computer Science



B.Sc. Hons. (CA&IT) / Data Science Sem-I
[U11E6CMT] COMPUTER MAINTENANCE
AND TROUBLESHOOTING

UNIT-5

Exploring Computer's Potential

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1. System Optimization Techniques

Improving computer performance without changing hardware.

♦ 1.1 Memory Management (RAM Handling)

- RAM = Temporary memory used while working on files or apps.
- Low RAM = Slow system
- Tips to manage memory:
 - Close unused programs and browser tabs.
 - Restart computer to clear memory.
 - Use Task Manager to check memory usage.
 - Shortcut: Ctrl + Shift + Esc

♦ 1.2 Managing Startup Programs

- Too many programs start when Windows starts = Slow boot.
- How to fix:
 - Open Task Manager → Startup tab.
 - Disable unnecessary apps (Right-click → Disable).
 - Keep antivirus and system tools enabled.

♦ 1.3 Clearing Temporary Files

- Temp files = cache, log files, recycle bin items.
- Clean them regularly:
 - Press Windows Key → Search "Disk Cleanup" → Select Drive (C:) → Clean temp files.
 - Improves speed and saves storage.

♦ 1.4 Disable Unnecessary Visual Effects

- Animations make system look fancy but use memory.
- For old PCs:
 - Right-click This PC → Properties → Advanced System Settings → Performance → "Adjust for best performance".

2. Upgrading Common Components (For Better Performance)

Sometimes hardware upgrades are needed to make the computer faster.

♦ 2.1 Upgrading RAM

- More RAM = Better multitasking.
- Example:

- 4GB RAM → Basic use.
 - 8GB RAM → Better for multitasking, development.
- Check compatibility before buying.

♦ 2.2 Upgrading Storage (HDD → SSD)

HDD	SSD
Slower	Faster
Noisy	Silent
Cheap	Slightly expensive
Moving parts	No moving parts (more reliable)

SSD = Faster boot, faster app loading, better performance.

♦ 2.3 External Storage (Backup or Extra Space)

- Use External Hard Disk, Pen Drive, or Cloud Storage.
- Frees up space from main drive.

3. Advanced Troubleshooting Resources

When something goes wrong, these can help.

♦ 3.1 Online Forums

- Free help from other users.
- Popular Forums:
 - Microsoft Community
 - Reddit (r/techsupport)
 - Tom's Hardware
 - Stack Overflow (for coding help)

Tips:

- Search your error message first.
- Be polite and clear when asking questions.

♦ 3.2 Manufacturer Support

- Visit official website of your device brand (HP, Dell, Lenovo, etc.)
- You can:
 - Download drivers
 - Get live support or chat
 - Use warranty services

♦ 3.3 Using Windows Built-in Troubleshooters

- Go to:
Settings → System → Troubleshoot → Other Troubleshooters
 - Can fix issues like:
 - No sound
 - Network not working
 - Windows update errors
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4. Safe Practices for Seeking Help

When facing computer problems, it's okay to ask for help — but it's very important to stay safe and protect your personal information and system from threats.

1. Never Share Personal Information

- Do NOT share:
 - Passwords
 - Bank or Aadhaar details
 - OTPs (One-Time Passwords)
 - Personal photos or sensitive documents
- Even if someone claims to be “tech support,” always be cautious.

2. Avoid Unknown Remote Access

- Some people may ask to control your PC remotely.
- Allow remote access ONLY:
 - If you're talking to official company support (like HP, Dell).
 - Through trusted software (like TeamViewer or AnyDesk) — and only when needed.
- Never leave the remote session open.

3. Use Official Websites and Sources

- Always download drivers or support tools from:
 - The manufacturer's website (like www.hp.com, www.dell.com).
 - Trusted sources like Microsoft, Google, etc.
- Avoid third-party or fake support websites.

4. Use Antivirus While Troubleshooting

- If you are:
 - Downloading new tools
 - Visiting tech forums or unknown pages
- Make sure antivirus is active to block malicious files or links.

5. Be Cautious on Online Forums

- Online communities like Reddit, Microsoft Forums, or Stack Overflow are helpful.

- But:
 - Don't follow risky instructions without understanding them.
 - Avoid clicking unknown links posted by users.
 - Verify the solution or ask a teacher/technician before doing something big.

6. Report Suspicious Activity

- If someone asks for money or personal info in exchange for help, report them.
- Inform your teacher or supervisor if you're unsure.

Topic	What It Does	Tools / Tips
Memory Management	Frees up RAM	Task Manager
Startup Programs	Speeds up boot time	Disable in Task Manager
Temp File Cleanup	Frees space	Disk Cleanup
Upgrade RAM	Better multitasking	Add compatible RAM
Upgrade to SSD	Faster system	Replace HDD with SSD
Online Help	Get support	Reddit, Microsoft, etc.
Manufacturer Support	Drivers & Repairs	Visit official websites
Safety Online	Protect your system	Don't share private info